# DoRAemon's Secret Tool for an A+ in Deep Learning: Weight-Decomposed Fine-Tuning

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Cornell University CS 5782 / CS 4782 Intro to Deep Learning

## Introduction

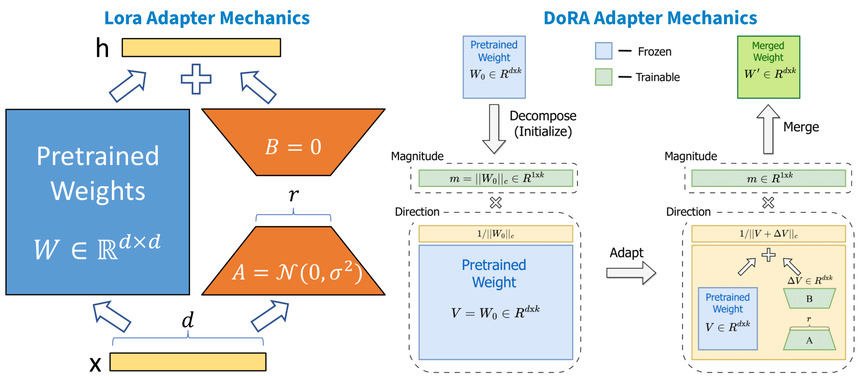
Full fine-tuning large language models is expensive. **LoRA** reduces the cost by training low-rank adapter updates while freezing the base model.[1]

**DoRA**

- **DoRA** decomposes pretrained weights into **magnitude** and **direction**.[3]
- Direction receives a **LoRA**-style low-rank update; magnitude is learned separately.

**Project goal**

- Does **DoRA** improve commonsense accuracy over a same-scope **LoRA** baseline?
- Reproduce the LLaMA2-7B trend and test attention-only, MLP-only, and full-scope adapters.




## Methodology

LLaMA

- We used 15k examples, not the full 170k, because compute was limited.
- Otherwise, we followed the **DoRA** paper and reported per-task plus macro-average accuracy.

**Controlled comparison.** Frozen base model, adapter-only checkpoints, 3 epochs, cutoff length 256, seed 42. Settings: **LoRA** r=32, LR=3e-4; rank-halved **DoRA** r=16, LR=2e-4.

## Our Independent Experiment: Scope Ablation

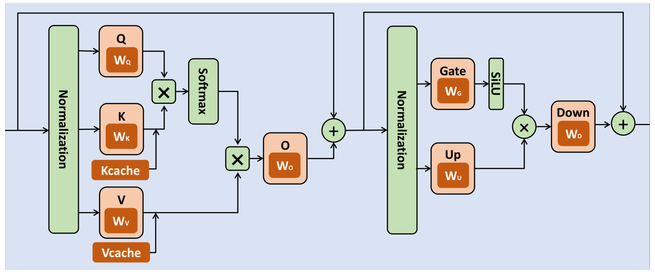



**Figure 1.** The ablation isolates adapter placement in attention projections, MLP projections, or both.

| Scope | Target modules | Purpose |
|---|---|---|
| **Full** | `q,k,v,up,down` projections | Paper-level reproduction |
| **Attention only** | `q,k,v` projections | Test routing/context layers |
| **MLP only** | `up,down` projections | Test feed-forward layers |

## Verdict: DoRA Improves Accuracy and Efficiency

| | | | | | |
|---|---|---|---|---|---|
| **Accuracy** | 77.41% | vs | 79.20% | +1.79 | ↑ |
| **Trainable Parameters** | 0.83% | vs | 0.43% | -48.2% | ↓ |

## The Adapter Duel

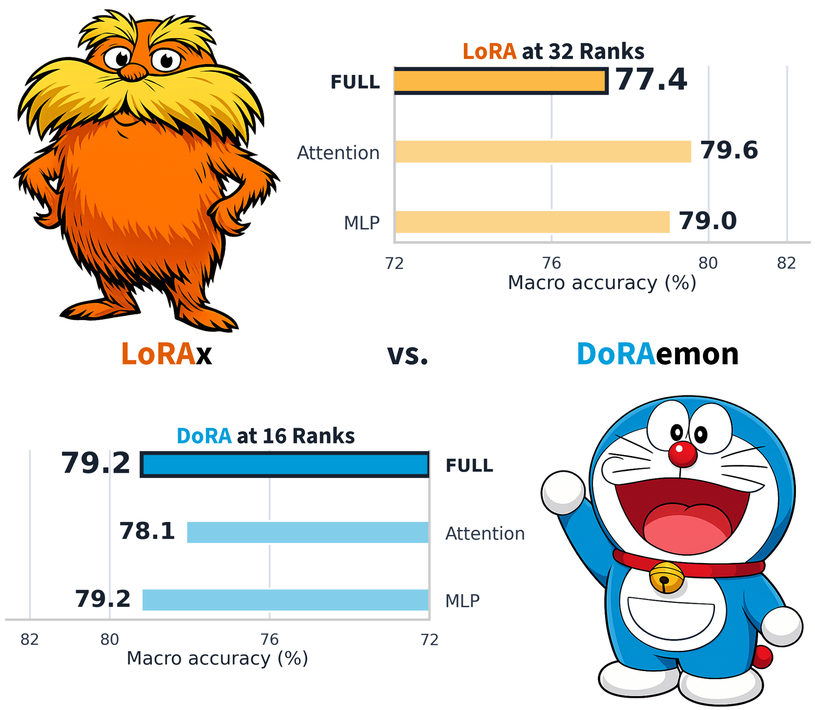



**Figure 2.** The full-scope bars are outlined: rank-halved **DoRA** improves macro accuracy from 77.4% to 79.2%.

## Benchmark with Largest DoRA Gain: HellaSwag (+6.0 pts)

### Task Format — Grounded Activity Completion

Given a short activity description, select the most plausible four-way continuation. Requires grounded physical-world commonsense.

**Context:** A woman is outside with a bucket and a dog. The dog is running around trying to avoid a bath. She pours soap on the dog and…

**(A)** lathers the dog up with a brush before rinsing.
**(B)** pours more water slowly into the bucket.
**(C) scrubs the dog down and rinses it thoroughly.**
**(D)** lets the dog run off and air dry on the lawn.

## Official Paper vs Full-Scope LLaMA2-7B Reproduction

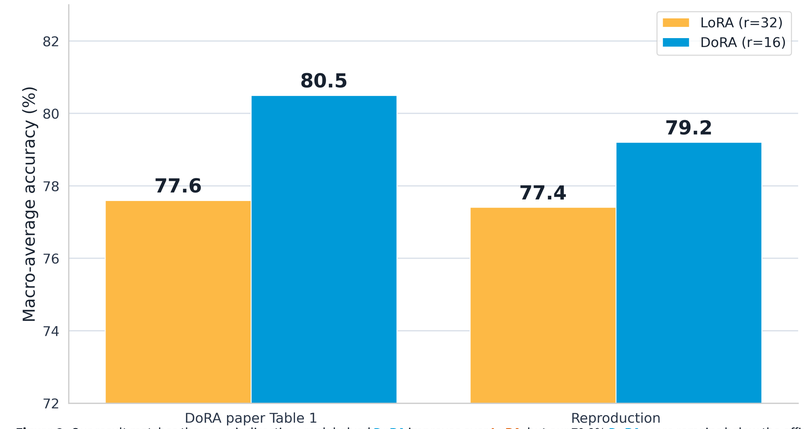



**Figure 3.** Our result matches the paper's direction: rank-halved **DoRA** improves over **LoRA**, but our 79.2% **DoRA** score remains below the official 80.5% reference.

## LoRA vs DoRA Across Ranks

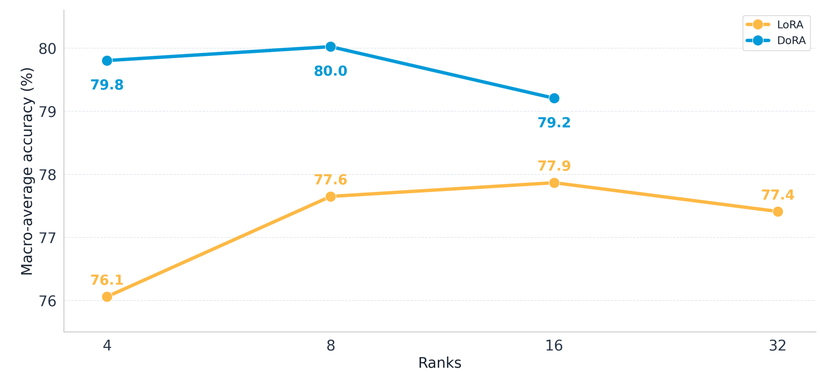



**Figure 4.** Macro-average accuracy across rank sizes, with the DoRA advantage shown in the lower panel.

## Conclusion & Future Work

### Takeaways

- Full-scope **DoRA** improved macro-average accuracy by **+1.79 points**.
- The reproduction supports the qualitative value of weight-decomposed low-rank adaptation.
- The ablation adds a caution: applying **DoRA** narrowly can remove the benefit.

### Next steps

- Train on the full `commonsense_170k` dataset.
- Repeat all six conditions across multiple random seeds.
- Add parameter-count, GPU-memory, and wall-clock cost comparisons.

## References

[1] Edward J. Hu et al. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.
[2] Hugging Face. Transformers documentation. `https://huggingface.co/docs/transformers`, 2026.
[3] Shih-Yang Liu et al. DoRA: Weight-decomposed low-rank adaptation. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. Oral presentation.
[4] NVIDIA Research. NVlabs/DoRA: Official implementation. `https://github.com/NVlabs/DoRA`, 2024.